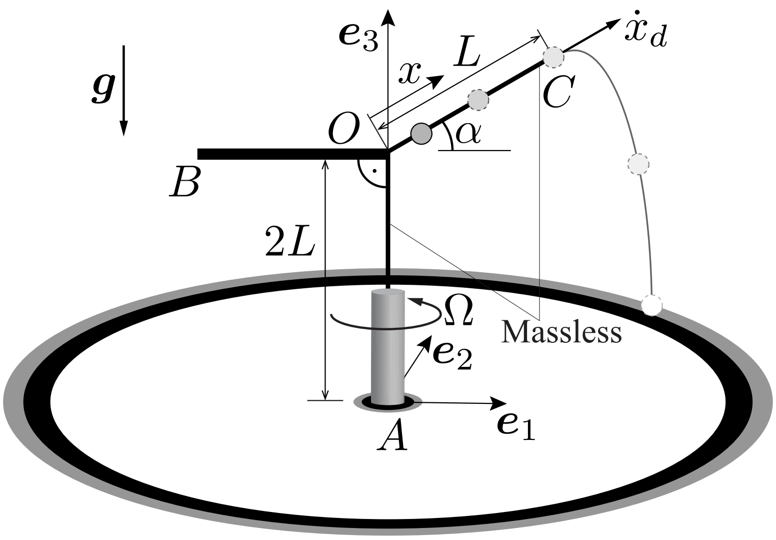
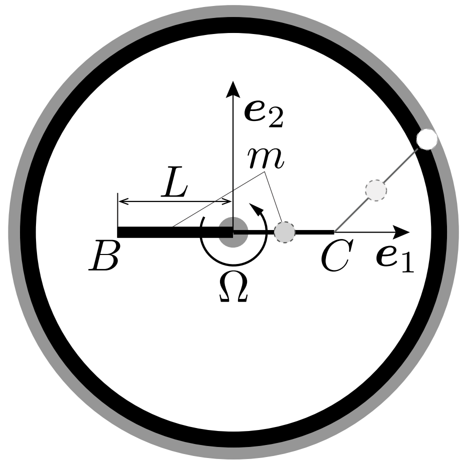
Problem 1 (60 Points)

A frame $ABCO$, consisting of two massless rigid rods AO and OC of lengths $2L$ and L, respectively, as well as slender rigid rod BO of mass m and length L, is rotating about the vertical $\boldsymbol{e}_3$-axis at a constant angular velocity $\boldsymbol{\Omega} = \Omega \boldsymbol{e}_3$. Due to the rotation of the frame, a particle of mass m slides frictionlessly up the rod OC, whose angle with respect to the horizontal direction is $\alpha = \pi/6$, as shown. Gravity acts down, as shown.

Given: m, L, $\alpha = \frac{\pi}{6}$, Ω, g, $\dot{x}_d = 4\sqrt{gL}$, γ





Top view

(a) Which of the following correctly describes the form of the moment of inertia tensor $\boldsymbol{I}_O$ of the system (including the particle and the frame) with respect to point O in the shown $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$-frame? Assume that $A \neq B \neq C \neq D \neq E \neq F$ and $A, B, C, D, E, F \neq 0$. (15 points)

(a) $[\boldsymbol{I}_\text{O}] = \begin{pmatrix} A & 0 & B \\ 0 & C & 0 \\ B & 0 & D \end{pmatrix}.$

(b) $[\boldsymbol{I}_\text{O}] = \begin{pmatrix} A & 0 & 0 \\ 0 & B & C \\ 0 & C & D \end{pmatrix}.$

(c) $[\boldsymbol{I}_\text{O}] = \begin{pmatrix} A & B & 0 \\ B & C & 0 \\ 0 & 0 & D \end{pmatrix}.$

(d) $[\boldsymbol{I}_\text{O}] = \begin{pmatrix} A & B & C \\ B & D & E \\ C & E & F \end{pmatrix}.$

(e) $[\boldsymbol{I}_\text{O}] = \begin{pmatrix} A & 0 & B \\ 0 & C & 0 \\ B & 0 & A \end{pmatrix}.$

(b) At what constant angular velocity Ω will the particle stabilize at the middle of the rod OC (i.e., remain at $x = L/2$)? (15 points)

From now on, assume that the particle reaches the end of the rod and falls off the frame at a known angular velocity $\Omega = \sqrt{\frac{g}{3L}}$ and with a speed along the rod of $\dot{x}_d = 4\sqrt{gL}$. Consider the $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$-frame *fixed* during the flight of the particle. Neglect air drag.

(c) At what time t_f after its departure from the rod does the particle hit the ground? (15 points)

(a) $t_f = 3\sqrt{\frac{L}{g}}$

(b) $t_f = 5\sqrt{\frac{L}{g}}$

(c) $t_f = 7\sqrt{\frac{L}{g}}$

(d) $t_f = \left(2\sqrt{3} + \sqrt{16 + \sqrt{3}}\right)\sqrt{\frac{L}{g}}$

(e) $t_f = \left(1 + \sqrt{6}\right)\sqrt{\frac{L}{g}}$

(d) Assuming that the particle hits the ground at $t_f = \gamma\sqrt{\frac{L}{g}}$, at what location in the shown (fixed) $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$-frame will the particle hit the ground? (15 points)

(a) $2\sqrt{3}\gamma L\boldsymbol{e}_1 + \frac{\gamma L}{2}\boldsymbol{e}_2$.

(b) $L\left(1 + 4\gamma\right)\boldsymbol{e}_1 + \frac{\gamma L}{\sqrt{3}}\boldsymbol{e}_2$.

(c) $\frac{\sqrt{3}L(1+4\gamma)}{2}\boldsymbol{e}_1$.

(d) $\frac{\sqrt{3}L(1+4\gamma)}{2}\boldsymbol{e}_1 + \frac{\gamma L}{2}\boldsymbol{e}_2$.

(e) $\frac{\gamma L}{\sqrt{3}}\boldsymbol{e}_2$.

Solution:

(a) Since the particle is at $x_1\boldsymbol{e}_1 + x_3\boldsymbol{e}_3$ with respect to point O, the form of it's moment of inertia is

$$[\boldsymbol{I}_O]_{\text{particle}} = m\begin{pmatrix} x_3^2 & 0 & -x_1x_3 \\ 0 & x_1^2 + x_3^2 & 0 \\ -x_1x_3 & 0 & x_1^2 \end{pmatrix} = \begin{pmatrix} A & 0 & B \\ 0 & \tilde{C} & 0 \\ B & 0 & \tilde{D} \end{pmatrix} \tag{1}$$

Next, the moment of inertia of a the rod along $\boldsymbol{e}_1$ is zero, and its center of mass is displaced only along this direction, so the form of the rod's moment of inertia is:

$$[\boldsymbol{I}_O]_{BO} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & E & 0 \\ 0 & 0 & E \end{pmatrix} \tag{2}$$

Adding up the two contribution leads to the following final form:

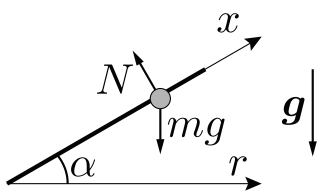
$$[\boldsymbol{I}_O] = \begin{pmatrix} A & 0 & B \\ 0 & C & 0 \\ B & 0 & D \end{pmatrix} \tag{3}$$

(b) Since the particle undergoes rotational motion, LMB in the radial direction is given by

$$m\left(\ddot{r} - r\Omega^2\right) = \Sigma F_r = -N\sin\alpha. \tag{4}$$

Here, r denotes the radial direction, and according to the free body diagram below $N = \frac{mg}{\cos\alpha}$ is the normal force applied by the bar (obtained from LMB in the direction normal to the bar, since the particle cannot move normal to the bar). Substituting the normal force expression into (4), and setting $r = \frac{L\cos\alpha}{2}$ and $\ddot{r} = 0$, yields

$$\Omega_{\text{mid}} = 2\sqrt{\frac{g}{3L}} \tag{5}$$



(c) Immediately after take off, the position and velocity of the freely flying particle in the vertical $\boldsymbol{e}_3$-direction are $x_3\,(t=0) = L\,(2+\sin\alpha)$ and $\dot{x}_3\,(t=0) = \dot{x}_d \sin\alpha$, respectively. Imposing these initial conditions in the vertical direction of LMB given by $m\ddot{x}_3 = -mg$ after integration yields the following expression of the vertical position of the particle's trajectory:

$$x_3\,(t) = -\frac{gt^2}{2} + \dot{x}_d \sin\alpha \cdot t + L\,(2+\sin\alpha) = -\frac{gt^2}{2} + 2\sqrt{gL}t + \frac{5L}{2}. \tag{6}$$

The time at which the particle hits the ground is found by setting $x_3\,(t_{\text{impact}}) = 0$, from which we obtain $t_f = 5\sqrt{L/g}$. Thus, the correct answer is (b).

(d) Since no forces are applied to the particle along the $\boldsymbol{e}_1$- and $\boldsymbol{e}_2$-directions, its 3D trajectory is given by

$$\begin{aligned}\boldsymbol{r} &= (L + \dot{x}_d t)\cos\alpha\,\boldsymbol{e}_1 + (\Omega L\cos\alpha)\,t\boldsymbol{e}_2 + x_3\boldsymbol{e}_3 \\ &= \frac{\sqrt{3}\left(L + 4\sqrt{gL}t\right)}{2}\boldsymbol{e}_1 + \frac{\sqrt{3}\Omega L}{2}t\boldsymbol{e}_2 + \left(-\frac{gt^2}{2} + 2\sqrt{gL}t + \frac{5L}{2}\right)\boldsymbol{e}_3.\end{aligned} \tag{7}$$

Substituting t_f, Ω yields the position at which the particle hits the ground:

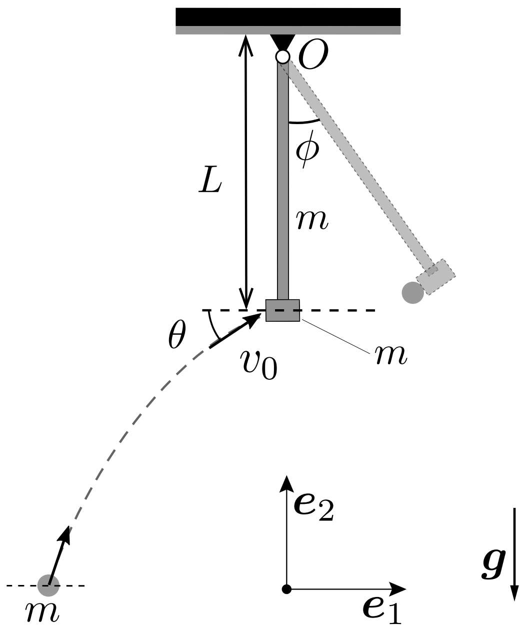
$$\boldsymbol{r}\,(t_f) = \frac{\sqrt{3}L\,(1+4\gamma)}{2}\boldsymbol{e}_1 + \frac{\gamma L}{2}\boldsymbol{e}_2. \tag{8}$$

The correct answer is (d)

Problem 2 (40 Points)

Consider a pendulum of total mass $2m$, consisting of a tip particle of mass m attached to a rigid rod of length L and mass m, which is hinged at point O, as shown below. The pendulum is initially at rest, hanging vertically down under the action of gravity. A particle of mass m is thrown from the ground, such that it collides with the pendulum at a velocity v_0 under an angle θ with the horizontal axis, as shown. After the collision, the particle sticks to the pendulum (e.g., there is some glue that makes it stick), and the entire system (pendulum with particle attached) rotates about point O. Assume that the problem is restricted to 2D.

Given: m, L, $\theta = \frac{\pi}{6}$, $v_0 = \sqrt{gL}$, g



(a) Assume that the particle has a speed $v_0 = \sqrt{gL}$ when it collides with the pendulum at an angle $\theta = \pi/6$. What is the maximum angle ϕ up to which the system swings after the collision, as shown? (20 points)

(b) Assume that the entire system (pendulum with the particle attached to it) is released from rest at an initial angle of $\phi = \arccos(1/15)$ to the vertical axis (i.e., initially $\cos\phi = 1/15$) . What is the magnitude of the reaction force $\boldsymbol{F}_O$ in the hinge at O, when the pendulum reaches its lowest point (vertical down)? (20 points)

Solution:

(a) To find the angular velocity of the system (ω_0) about point O just after the collision, conservation of the angular momentum of the system (particle + pendulum) about point O *during the collision* is exploited. (Note that angular momentum is not constant during the swinging of the pendulum, since gravity produces a torque with respect to point O; yet, during the collision – when the pendulum is vertical down – angular momentum is conserved) .To use angular momentum conservation, we must find the momentum of inertia of the system about the $\boldsymbol{e}_3$-axis at point O. This can be written as

$$I_{O,33} = 2mL^2 + \frac{1}{3}mL^2 = \frac{7}{3}mL^2. \tag{9}$$

Now, the initial angular momentum of the system about O is due to the moving particle and is equal to

$$H_O^{intial} = mv_0 L\cos\theta = \frac{\sqrt{3}}{2} mv_0 L. \tag{10}$$

The final angular momentum of the system about O is due to the swinging system and is equal to

$$H_O^{final} = I_{O,33}\omega_0 = \frac{7}{3} mL^2\omega_0. \tag{11}$$

Equating the two expressions above yields $\omega_0 = \frac{3\sqrt{3}v_0}{14L}$.

Now, to obtain the angle ϕ, conservation of the total kinetic and (gravitational) potential energy between the lowest and highest points of the pendulum trajectory can be used. The total kinetic energy T at the starting point of the pendulum can be written as a sum of the kinetic energy of rotation of the rod of mass m and the kinetic energy of the particles at the end of the rod.

$$T_{\text{initial}} = \frac{1}{2}\frac{1}{3}mL^2\omega_0^2 + \frac{1}{2}2m(L\omega_0)^2. \tag{12}$$

The potential energy V of the system at the highest point can be written as

$$V_{\text{final}} = mg\frac{L}{2}(1-\cos\phi) + 2mgL(1-\cos\phi). \tag{13}$$

Equating the two expressions, we obtain

$$\frac{7}{6}mL^2\omega_0^2 = 5mg\frac{L}{2}(1-\cos\phi). \tag{14}$$

Substituting the expression for ω_0, and using $v_0 = \sqrt{gL}$, one finds $\cos\phi = \frac{131}{140}$. Hence, the maximum angle of swing for the pendulum is

$$\phi = \cos^{-1}\frac{131}{140}. \tag{15}$$

(b) First, we need to find the angular velocity (ω) of the system when it is at the lowest point of its trajectory.Conservation of the total kinetic and potential energy can be used. Both energies have a component due to the particles at the end of the rod, and another due to the rod of mass m:

$$\begin{aligned} &2mgL(1-\cos\phi) + mg\frac{L}{2}(1-\cos\phi) = \frac{1}{2}2m(\omega L)^2 + \frac{1}{2}\frac{1}{3}mL^2\omega^2 \\ &\omega^2 = \frac{15g}{7L}(1-\cos\phi) \end{aligned} \tag{16}$$

At the lowest point of the trajectory, the system experiences a centripetal force:

$$\boldsymbol{F}^{\text{centripetal}} = \left(2m\omega^2 L + m\omega^2\frac{L}{2}\right)\boldsymbol{e}_2 = \frac{5}{2}m\omega^2 L\boldsymbol{e}_2. \tag{17}$$

Using LMB at this point of the pendulum's trajectory leads to

$$\begin{aligned} &\mathbf{N} - 3mg\boldsymbol{e}_2 = \mathbf{F}^{centripetal} \\ \Rightarrow\quad &\mathbf{N} = 3mg\left(1 + \frac{25}{14}(1-\cos\phi)\right)\boldsymbol{e}_2 \end{aligned} \tag{18}$$

Using the information that $\cos\phi = 1/15$ finally gives

$$\mathbf{N} = 8mg\,\boldsymbol{e}_2. \tag{19}$$